

THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY  
**ECON 4999B: Quantitative Macroeconomics with AI  
Integration**

Instructor: Prof. Yang LU

Date: Dec 10, 2025

Time allowed: 120 minutes

**Not to be taken away.**

**Instructions:**

- Answer the questions as instructed. Write your answers on the answer book. Anything written on the question book will NOT be graded.
- Write your answer to all the questions within the provided area. **Anything outside the provided area will NOT be graded.**
- Make sure that all your handwriting is legible. Anything that cannot be understood by the grader will not be graded.
- Please submit BOTH the question book and the answer book after the exam.

**DO NOT OPEN UNTIL INSTRUCTED!**

**Name:**

**Student ID:**

**Group Number:**

## **Part I: Research Methodologies (60 points)**

Choose 4 out of the following () topics which are all based on your classmates' presentations. Each part carries 15 points. Please specify which questions you are answering.

**You MUST choose the topic presented by your group.** If you answer more than 4 topics, only the first 4 will be graded.

## **Part II: Research Workflow (15 points)**

Read the following report of a student research. Answer the questions based on the report.

## **Part III: Research Design (25 points)**

Read the following background knowledge and the research question. Answer the questions on how to design the research.

**\*\*\*\*\* END OF THE EXAM \*\*\*\*\***